

Common Issues in Prescribing in Kidney Disease and Kidney Replacement Therapy

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Appropriate prescribing for people with kidney disease is underpinned by the principles discussed in [Chapter 77](#). Even experienced physicians will frequently need to check reference sources to be sure of the correct prescribing approach to medications with which they are less familiar. Up-to-date reference databases should be used to inform prescribing in patients with chronic kidney disease (CKD), those on dialysis, and after kidney transplantation.¹⁻⁴

ANALGESICS

Various analgesics are excreted by the kidney, and dose modification or avoidance is required in people with impaired kidney function.⁵ Fear of adverse effects often prevents use of sufficient doses.

Acetaminophen (Paracetamol)

Despite suggestions that acetaminophen is nephrotoxic and has modest analgesic potency, its lack of platelet inhibition and gastrointestinal toxicity make it a safer base analgesic in CKD and renal replacement therapy than nonsteroidal antiinflammatory drugs (NSAIDs) and opioids. Acetaminophen is hepatically metabolized and does not require dose adjustment in dialysis or transplant patients.

Opioid Analgesics

Several opioids have active/toxic metabolites that are removed by the kidney,⁶ and accumulation of these metabolites prolongs drug action and predisposes to central nervous system (CNS) toxicity (sedation, respiratory depression, confusion, hallucinations, and seizures). Regular opioid use in fluid-restricted kidney patients can exacerbate constipation. Opioids should be used cautiously in CKD, with adequate doses given to establish control and titrated to the smallest effective dose. Morphine is metabolized to two active metabolites (morphine 3-glucuronide and morphine 6-glucuronide) that can accumulate in CKD, causing CNS toxicity, respiratory depression, and sedation. Meperidine (pethidine) should be avoided in moderate to severe CKD because it is metabolized to a metabolite (normeperidine) that accumulates in CKD and can cause CNS toxicity (seizures, myoclonus, opisthotonus, mental state changes, respiratory depression, and psychosis).⁷ Dextropropoxyphene should be avoided because it is metabolized to the metabolite norpropoxyphene, which is removed by the kidney and can cause cardiac toxicity.

Hydromorphone is metabolized to hydromorphone 3-glucuronide, which has minor activity and does not accumulate substantially. Buprenorphine is metabolized to relatively inactive metabolites which are excreted in bile, and it is therefore relatively safe. Weaker opioids (codeine, dihydrocodeine, and hydrocodone) can still cause CNS and respiratory depression. With appropriate titration, alfentanil, fentanyl, sufentanil, methadone, and oxycodone are relatively safer choices because their metabolites are largely inactive. For this reason, fentanyl

(parenteral and topical) and oxycodone (oral) are usually preferred opioids in CKD. The partial opioid tramadol is metabolized to an active metabolite, O-desmethyltramadol, whose half-life doubles in CKD and predisposes to seizures, respiratory depression, and other CNS adverse effects. In opioid intoxication, naloxone may be used at normal dose for reversal.

Nonsteroidal Antiinflammatory Drugs

The most frequent nephrotoxic effect of NSAIDs is acute kidney injury (AKI) caused by prevention of prostaglandin-mediated, afferent arteriolar vasodilation (see Chapter 70). In healthy individuals, cyclooxygenase (COX) inhibition has little effect on kidney function, but in individuals with CKD, nephrotoxicity is more likely to occur due to the greater dependence on prostaglandins to maintain kidney perfusion. Combining NSAIDs with angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), diuretics, or other antihypertensives increases the potential for nephrotoxicity, and such combinations should be prescribed cautiously.⁸ If essential, in stable patients with CKD, NSAIDs should be used at the lowest effective dose for the shortest period with monitoring of kidney function. Increases in plasma potassium and sodium, as well as fluid retention, also may occur. NSAIDs (especially fenoprofen) can cause a rare idiosyncratic reaction that may present with acute interstitial nephritis or nephrotic syndrome with a glomerular lesion resembling minimal change disease. Because COX-2 is expressed in the kidney, COX-2-selective agents offer no advantage in kidney toxicity over nonselective inhibitors.⁹ The potential for cardiovascular (CV) complications of COX-2 inhibitors is also undesirable in kidney patients at high risk of CV disease.

Drugs for Neuropathic Pain

Low-dose tricyclic antidepressants are used at normal doses in patients with reduced glomerular filtration rate (GFR), although they may be more sensitive to anticholinergic side effects. Low-dose valproate and carbamazepine are used at normal doses. Extreme caution and dose reductions are required with gabapentin¹⁰ and pregabalin¹¹ because they are extensively removed by the kidney and can cause significant CNS adverse effects (somnolence, lethargy, dizziness, myoclonus, and ataxia).

ANTIMICROBIAL AGENTS

The kidneys excrete many antiinfective agents, and dose reduction is often required in severe CKD or those undergoing extended therapy.¹² Many other antiinfectives have a wide therapeutic window, and dose adjustment is often clinically unnecessary despite their reliance on kidney excretion. An important principle of antiinfective therapy is to initiate effective drugs at sufficient doses rather than using conservative dose reductions that fail to achieve effective drug concentrations.

Normal or larger than expected doses based on kidney function may be appropriate when treating less susceptible organisms or when drug distribution to the site of infection is reduced (e.g., meningitis and urinary tract infections [UTIs]). Severe infection, especially in severely immunocompromised patients, may require extended therapy. In patients with UTI and severe CKD (e.g., $\text{GFR} < 30 \text{ mL/min/1.73 m}^2$), anti-infectives that are only filtered by the kidneys are unlikely to achieve adequate concentrations in the urinary tract and are therefore unlikely to result in clinical cure. Because plasma concentrations will also be elevated, systemic toxicity is more likely (e.g., trimethoprim and nitrofurantoin). Administering a reduced dose of such anti-infectives is illogical because concentrations in the urinary tract will be further decreased and efficacy further reduced. Anti-infectives that are secreted by the tubules in addition to being filtered (e.g., penicillins) are more likely to achieve adequate concentrations in the urinary tract in advanced CKD and preferred agents.

Antibacterials

Aminoglycosides

Aminoglycosides are excreted unchanged by the kidneys and can cause nephrotoxicity, vestibular toxicity, and ototoxicity that can be severe and irreversible.¹³ When use is required, the total daily dose should be reduced, treatment courses kept to a minimum, and serum concentrations and kidney function monitored. Because aminoglycosides achieve high urine concentrations even in advanced CKD, lower doses may be appropriate for uncomplicated UTIs.

Aminoglycoside dosage schedules. Despite short half-lives (2–3 hours in normal kidney function), aminoglycosides are usually given once daily rather than in divided daily doses. This is based on two properties, namely, concentration-dependent killing and postantibiotic effect. The former requires that a high peak concentration (ideally 4–5 times the pathogen minimum inhibitory concentration [MIC]) is achieved. The latter allows the plasma concentration to fall below the MIC for much of the dosing interval without compromising efficacy. This “larger dose given less frequently” approach has been shown to be more efficacious and less toxic. The required dose regimen depends on volume of distribution (V_D) and kidney clearance. Determining the best dose regimen in CKD involves reducing the normal daily dose (5–7 mg/kg/day)¹⁴ for gentamicin/tobramycin or 15 mg/kg for amikacin in proportion to kidney function (e.g., with 25% remaining kidney function, corresponding to GFR of $\sim 25 \text{ mL/min/1.73 m}^2$ or creatinine clearance [CrCl] of 25 mL/min , reduce total daily dose to 25% of normal). Aminoglycosides should be administered at a minimum effective dose ($\geq 3 \text{ mg/kg}$ for gentamicin and tobramycin; $\geq 7.5 \text{ mg/kg}$ for amikacin) to achieve effective peak serum concentrations ($> 10 \mu\text{g/mL}$ for gentamicin and tobramycin; $> 30 \mu\text{g/mL}$ for amikacin). It is often necessary to increase the interval between doses (to > 24 hours) to allow for an adequate dose (that will achieve a good peak concentration) to be administered.

Aminoglycoside concentration monitoring. Because of toxicity and pharmacokinetic variability in patients with infection and CKD, monitoring is essential to guide aminoglycoside dosing, particularly when therapy is initiated or significant changes occur in kidney function or clinical condition. Routine monitoring is not required for short treatment (< 48 hours). Graphic methods using a single-point concentration 6 to 14 hours after administration are valid only with relatively normal GFR . They do not provide information on attainment of peak concentrations (for efficacy) and are not recommended in patients with CKD. Trough concentrations can be measured but must be interpreted in the context of the time after the dose when the sample was drawn. Trough monitoring does not provide information on the attainment of peak concentrations. Some patients, especially

those with severe sepsis and CKD, may have altered (usually higher) V_D , which results in lower peak concentrations. Measuring peak concentrations (30 minutes after dosing) is the only certain way to determine dose adequacy. Combining peak and trough monitoring (approximately two half-lives after dosing) provides information on both V_D and clearance and can therefore be used to determine both the size of individual doses and the administration interval. Provided the relationship between the time of administration and blood sampling is known, simple dose modifications are made on first principle estimates (see Chapter 77). Peak and trough target levels for aminoglycosides are shown in Table 77.6.

Carbapenems. Doripenem, imipenem, and meropenem are significantly excreted in urine but with appropriate dose modification can be used safely. Imipenem can cause significant neurotoxicity (myoclonic activity, seizures, and confusion), especially in high doses in those with underlying CNS disorders or CKD. Imipenem is inactivated by dehydropeptidase 1 in the kidney and is combined with cilastatin to prevent this.

Cephalosporins. Although most cephalosporins are predominantly removed by the kidney, their relative safety means that normal doses are used for most short-course therapies even in dialysis patients (e.g., ceftriaxone, cefaclor, and cephalexin). High-dose parenteral therapy and prolonged courses of some agents require dose reduction in severe impairment (e.g., cefepime, cefotaxime, ceftazidime, cefoxitin, and cefazolin). Therapeutic concentrations of cefazolin are maintained after doses of 20 mg/kg after dialysis three times weekly.¹⁵ Some cephalosporins (e.g., cefoxitin) interact with colorimetric assays for creatinine, resulting in falsely elevated serum creatinine values.¹⁶

Fluoroquinolones. Kidney excretion is significant with ciprofloxacin, norfloxacin, and gatifloxacin, and doses should be reduced if GFR is less than $30 \text{ mL/min/1.73 m}^2$.¹⁷ Moxifloxacin is less dependent on kidney excretion, and dose reduction is not required. Norfloxacin plasma concentrations are too low for treating bloodborne infections, but urinary concentrations are adequate for treating UTIs. Quinolones are generally well tolerated but can cause CNS effects (headache, dizziness, insomnia, depression, restlessness, and tremors), interstitial nephritis, musculoskeletal effects (tendonitis and arthralgias), aortic aneurysms, and crystalluria. If use is essential, quinolones should be administered at least 1 hour before or 2 hours after meals and metallic phosphate binders, which may reduce absorption.

Glycopeptides. Vancomycin is a glycopeptide that is extensively removed by the kidney. Because of nephrotoxicity and ototoxicity, dose modification is essential even in mild to moderate impairment. Nephrotoxicity is more likely in those with preexisting CKD, with prolonged therapy, high plasma concentrations, and when concomitant nephrotoxins are prescribed. Ototoxicity may involve sensorineural deafness and tinnitus and is more likely with previous auditory compromise or impaired GFR . Rapid infusion also can cause an erythematous anaphylactoid-like reaction involving the upper body (red man syndrome) that usually resolves with slowing of the infusion rate and antihistamines. If vancomycin use is essential, duration of therapy should be minimized with regular monitoring of serum concentrations and GFR . Vancomycin should be administered once plasma serum concentrations have fallen below target values that are related to the MIC. Vancomycin distributes into body tissues, and clearance from the bloodstream after initial doses can appear greater than expected from estimates of GFR or CrCl . Once these tissues are saturated, clearance is purely via kidney clearance and is relatively predictable. In dialysis patients, single doses (e.g., 1 g of vancomycin) can maintain therapeutic concentrations ($> 15 \mu\text{g/mL}$) for 3 to 5 days.¹⁸ Vancomycin is removed more extensively with modern high-flux than with older low-flux dialysis membranes. For teicoplanin, dose reduction is necessary when CrCl is less than 50 to 60 mL/min .

Lincosamides. Clindamycin and lincomycin are relatively safe and are not significantly excreted by the kidneys.

Macrolides. Macrolides are mostly hepatically cleared, and dose modifications are usually not required even in patients with advanced kidney failure. Dose reduction is required with high-dose or intravenous erythromycin, which may prolong QT interval and cause ototoxicity. Various macrolides are potent inhibitors of cytochrome P450 CYP3A4, 1A2, and P-glycoprotein, so there may be major interactions with and increased exposure to coadministered drugs that rely on CYP3A4, 1A2, or P-glycoprotein-mediated absorption and metabolism (including cyclosporine, tacrolimus, sirolimus, everolimus, and statins). Erythromycin and clarithromycin are the most potent inhibitors. Roxithromycin and azithromycin are much weaker inhibitors, causing few or no significant interactions (see Table 78.2).

Penicillins. Most penicillins have a short half-life and are rapidly eliminated by glomerular filtration and tubular secretion. Most have a relatively wide therapeutic window and are used at normal doses for short courses of oral therapy (e.g., amoxicillin) regardless of kidney function. High-dose parenteral therapy or prolonged high-dose oral therapy may require dose reduction to prevent electrolyte disturbances and neurotoxicity in kidney failure. Because the bactericidal effect of penicillins is not dependent on antibiotic concentration and exerts little or no postantibiotic effect, the amount of time above the MIC is more important than the maximum concentration. This translates into the frequency of dosing being more important than the size of each dose. For patients with UTIs who also have CrCl less than 40 to 50 mL/min, penicillins are a good choice (if the pathogen is sensitive) because urinary concentrations are higher than with agents that are merely filtered (e.g., trimethoprim). Dose reduction of benzyl penicillin, ticarcillin-clavulanate, and piperacillin-tazobactam are advised in patients with CrCl less than 40 to 50 mL/min.

Rifamycins. Rifamycins are mainly hepatically metabolized and used at normal doses. Orange-red coloration of body fluids (e.g., urine and peritoneal dialysis [PD] fluid) is common. These agents cause hepatotoxicity and blood dyscrasias. Rifampin is a potent CYP450 enzyme inducer and significantly increases metabolism of drugs, including immunosuppressants and oral contraceptives. Rifabutin does not induce CYP450 enzymes to the same extent.

Tetracyclines. Tetracycline was the originator of this class but has largely been replaced by newer derivatives due to the need for frequent administration, adverse effects (including Fanconi-like syndrome from outdated tetracycline), and resistance among pathogens. In CKD, tetracycline should be avoided because it is catabolic, causing uremia, hyperphosphatemia, and metabolic acidosis; it also may aggravate preexisting kidney failure.¹⁹ Doxycycline, minocycline, and tigecycline are newer derivatives that are not significantly dependent on kidney clearance and may be used as usual.

Sulfonamides and trimethoprim. Many sulfonamides are eliminated by acetylation followed by kidney excretion. Acetylated metabolites may cause crystalluria and tubular damage. Sulfamethoxazole and trimethoprim display similar kidney excretion except at extremes of urine pH. Alkaline urine promotes sulfamethoxazole excretion, whereas acidic urine promotes trimethoprim excretion. Both accumulate in CKD, and dose reduction is advisable except perhaps in the treatment of UTIs when higher plasma concentrations might be required to facilitate adequate urinary concentrations. Trimethoprim inhibits tubular secretion of creatinine, resulting in an increase in serum creatinine²⁰ that does not reflect a true change in GFR and can be misinterpreted as nephrotoxicity.

Other antibiotics. Linezolid is used at a normal dose. Metronidazole has a partially active metabolite, although only 15% of the parent drug is removed by the kidney. It is usually given in usual doses or reduced to twice-daily doses with dialysis.

Antimycobacterials

Ethambutol is extensively removed by the kidney, and dose reduction is required when CrCl is less than 30 mL/min to avoid visual toxicity.²¹ Isoniazid, pyrazinamide, and rifampin-rifabutin can be given in usual doses. Rifampin causes induction of hepatic enzymes and severe drug interactions with transplant immunosuppressants. Amikacin, kanamycin, streptomycin, capreomycin, and gatifloxacin have extensive kidney clearance, are nephrotoxic, and require significant dose reduction.

Antifungals

Amphotericin. Amphotericin use is limited by nephrotoxicity (see Chapter 55). Nephrotoxicity with crystalline amphotericin may be minimized with prehydration and administration by continuous infusion. Liposomal formulations have fewer infusion-related problems and are less nephrotoxic but more expensive. Oral amphotericin (used for its enteric effect and lozenges for oral thrush) is not absorbed and does not contribute to nephrotoxicity.

Azole antifungals. Most azole antifungals (ketoconazole, itraconazole, posaconazole, and voriconazole) are extensively metabolized and do not require dose reduction. Fluconazole is excreted by the kidney; after a loading dose, maintenance doses should be reduced in moderate to severe CKD, and it should be given after dialysis. Fluconazole is significantly excreted in urine and hence is often useful for fungal UTIs. The manufacturer recommends avoiding intravenous voriconazole in severe CKD because of possible accumulation of the intravenous vehicle (sulfobutyl betadex sodium). Voriconazole, ketoconazole, and itraconazole are potent inhibitors of CYP3A4 and P-glycoprotein that are involved in the metabolism and absorption of a variety of drugs, including cyclosporine, tacrolimus, sirolimus, everolimus, and statins. A twofold to 10-fold reduction of the concomitant drug may be required (guided by drug monitoring; see Table 78.2). Interactions between ketoconazole and calcineurin inhibitors (CNIs) or mammalian target of rapamycin (mTOR) inhibitors have been exploited as a cost-reducing immunosuppressant-sparing strategy. Increased exposure to statins (except pravastatin) caused by CYP3A4 inhibition increases the risk for rhabdomyolysis-induced CKD. Fluconazole is a weak enzyme inhibitor, and although caution should be exercised, the need for preemptive dose reduction of concomitant drugs is less certain. Topical azoles, including bifonazole, clotrimazole, econazole, ketoconazole, tinidazole, and miconazole, are minimally absorbed and do not cause interactions.

Other antifungals. Absorption of nystatin from topical and oral preparations is minimal and safe in CKD. Terbinafine is hepatotoxic, and a 50% dose reduction is suggested in moderate to severe CKD. Griseofulvin, anidulafungin, and caspofungin can be given in usual doses. Flucytosine has extensive kidney clearance and requires significant dose reduction.

ANTIVIRALS

Many antivirals (or their active metabolites) have extensive kidney excretion and can cause nephrotoxicity (see also Chapter 70).

Guanine Analogs

Acyclovir, its prodrug valacyclovir, and famciclovir (a prodrug of penciclovir) have extensive kidney clearance, and doses must be reduced in proportion to the severity of CKD. These agents can also crystallize in tubules, causing obstructive nephropathy, especially when given intravenously. Where possible they should not be administered in low volumes of fluid, and the patient should be well hydrated. High concentrations cause severe CNS toxicity (cerebral irritation, ataxia, and myoclonus). Ganciclovir and its prodrug valganciclovir have extensive

kidney clearance,²² and accumulation leads to severe bone marrow toxicity. Dose reduction based on kidney function is essential. These agents are freely dialyzed and should be given after hemodialysis (HD).

Hepatitis B and C

Pegylated interferon alfa-2a has a larger metabolic clearance than pegylated interferon alfa-2b, which has higher kidney clearance and requires dose reduction in CKD. Interferons can upregulate cell surface expression of class II histocompatibility antigens, which increases the potential for transplant rejection. Ribavirin and its metabolites rely on kidney excretion; accumulation causes severe anemia, making it contraindicated if GFR is below 50 mL/min; significantly reduced doses are sometimes used in lower levels of GFR. Newer direct-acting antivirals for hepatitis C (boceprevir, daclatasvir, ledipasvir, sofosbuvir, ombitasvir, paritaprevir, ritonavir, dasabuvir, simeprevir, telaprevir) are less dependent on kidney excretion and may be preferable in patients with CKD. Adefovir, entecavir, lamivudine, and telbivudine have extensive kidney excretion, and significant dose reduction is essential.

Neuraminidase Inhibitors

Oseltamivir is converted by hepatic esterases to its active metabolite, oseltamivir carboxylate, which is extensively (99%) removed through glomerular filtration and tubular secretion.²³ The area under the curve (AUC) is increased 10-fold in severe CKD, and, although the drug is well tolerated, dose reduction is recommended. Oseltamivir carboxylate is cleared by hemodialysis, but in anuric patients plasma concentrations remain virtually unchanged in between hemodialysis sessions. Patients on hemodialysis who have even small amounts of residual kidney function can clear oseltamivir carboxylate significantly faster than those who are anuric. Ideally, dosing in CKD would be guided by therapeutic drug monitoring, but this is not routinely available. Reducing the dose of oseltamivir in CKD used to be achieved by giving the same dose (75 mg) less frequently, but more recently it has been suggested it should be achieved by giving a lower dose (30-mg capsule) once or twice a day, depending on the degree of CKD. Giving a smaller dose from the outset delays the attainment of critical “therapeutic concentrations” because it takes approximately four times the prolonged half-life to achieve concentrations to maximize benefit. The initial dose should be 75 mg, and subsequent doses can be reduced either by reducing the dose size or prolonging the interval between doses.²⁴ It may be wiser to use zanamivir whenever CrCl falls below 20 mL/min. Zanamivir is also removed by the kidney, but dose modification is not necessary because absorption from the inhaled formulation is very low (~2%).

Other Antivirals

Cidofovir and foscarnet have extensive kidney clearance and require dose reduction in CKD. Both are nephrotoxic, require close monitoring of kidney function, and should be administered with hydration. Cidofovir is recommended to be administered with probenecid to slow kidney secretion and minimize nephrotoxicity. Tenofovir doses should also be reduced in moderate chronic kidney failure and avoided below 30 mL/min/1.73 m² and in combination with other nephrotoxins, such as adefovir. Agents active against HIV infection are also discussed in Chapter 58.

IMMUNOSUPPRESSANTS

Dosing of immunosuppressive agents used in kidney transplantation, as well as adverse effects, are discussed in Chapter 106. Initial and maintenance doses in individuals are highly variable and depend on

local protocol, concomitant therapy, rejection risk, drug concentrations, and response. Immunosuppression is required for the life of the transplant and should never be withheld except in exceptional circumstances (life-threatening infection or malignancy). Regimens should preferably be convenient (e.g., twice daily) to enhance compliance. In practice, the precise timing of doses or intake relative to time and food is not as important as consistency in relation to time and food because doses can be titrated to levels and response. Most immunosuppressants are hepatically metabolized and do not require dose modification based on altered excretion in CKD.

Immunosuppressants are prone to a range of significant drug interactions, which should be considered whenever a patient receiving these agents has a change in drug regimen. Significant interactions also can occur with “natural” or “herbal” medicines (e.g., St. John’s wort; see also Chapter 77 and Table 78.1). If these products are deemed essential, it is advisable to monitor response, kidney function, blood counts, and serum drug concentrations regularly but especially during initiation, dose alterations, and stopping the “natural” product. Therapeutic monitoring is recommended for several immunosuppressants²⁵ (tacrolimus, cyclosporine, everolimus, sirolimus, and mycophenolate). The target range may differ from center to center due to assay differences and may change with time posttransplant and immunosuppressive regimen. Reported concentrations should always be interpreted in the context of patient response and adverse effects.²⁶

Calcineurin Inhibitors

CNIs are extensively metabolized to inactive metabolites. A major adverse effect of CNIs is nephrotoxicity (acute and chronic),²⁷ and regular assessment of kidney function is essential. Nephrotoxicity may increase with concurrent use of mTOR inhibitors. Early detection of CNI nephrotoxicity on biopsy assists in planning maintenance therapy (see Chapters 106 and 109). Lower doses or temporary withdrawal may be used when CNIs have caused or are likely to cause nephrotoxicity (e.g., delayed graft function). Calcium channel blockers (CCBs) and blockers of the renin-angiotensin system may both reduce CNI nephrotoxicity. CNIs are substrates of P-glycoprotein and CYP3A4. Drugs that inhibit or compete as substrates for these can increase absorption and reduce metabolism of CNIs, causing increased serum concentrations and adverse effects. Drugs that induce these can reduce absorption and increase metabolism of CNIs, causing reduced serum concentrations, and increase the risk for rejection (Table 78.2). Tacrolimus trough concentrations (C₀) show good correlation with drug effect/toxicity and are used to monitor therapy. Clinically, a combination of C₀ (trough), C₂ (concentration 2 hours after dose), and AUC₀₋₄ (multipoint AUC, 0–4 hours) has been used to measure cyclosporin. Different assay methodologies (high-performance liquid chromatography [HPLC] and immunoassays) have differing abilities to differentiate parent compound from inactive metabolites. Specific HPLC methods are preferred, and physicians should be aware that results from different laboratories may differ due to assay specificity.

Corticosteroids

Corticosteroids are predominantly cleared by hepatic metabolism to inactive metabolites. Dose modification based on kidney function is not required.

Antiproliferative and Cytotoxic Agents

Various antiproliferative and cytotoxic agents are used to prevent transplant rejection (azathioprine and mycophenolate) and to suppress the immune system in patients with autoimmune and inflammatory kidney disorders (azathioprine, cyclophosphamide, chlorambucil, methotrexate, and mycophenolate). Their primary dose-limiting toxicity is bone marrow

TABLE 78.1 Common Drug Dosing Recommendations in Chronic Kidney Disease Including Kidney Transplantation and Dialysis

	PREDOMINANT ELIMINATION PATHWAY			
Drugs	Hepatic	Kidney	Dosing Recommendations	Comments
Analgesics				
Simple analgesics	Acetaminophen		Normal dosing	
Opioids		Morphine	Avoid/dose reduction	Metabolite kidney clearance
		Pethidine (meperidine)	Avoid	Metabolite kidney clearance
	Fentanyl		Cautious normal doses	
	Hydromorphone		Cautious normal doses	
	Methadone		Cautious normal doses	
	Oxycodone		Cautious normal doses	
	Tramadol		Cautious normal doses	
NSAIDs			Avoid if possible	Potential nephrotoxicity
Neuropathic pain		Gabapentin	Significant dose reduction	CNS adverse effects
		Pregabalin	Significant dose reduction	CNS adverse effects
	Carbamazepine		Normal dose	
	Low-dose TCAs		Normal dose	
	Valproate		Normal dose	
Antiinfectives				
Aminoglycosides		Amikacin	Avoid/dose reduction	Nephrotoxicity
		Gentamicin	Avoid/dose reduction	Nephrotoxicity
		Tobramycin	Avoid/dose reduction	Nephrotoxicity
Carbapenems		Ertapenem	Reduce in severe impairment	
		Doripenem	Dose reduction	
		Imipenem	Dose reduction	
Cephalosporins		Cefotaxime	Dose reduction	
		Ceftazidime	Dose reduction	
		Cephalexin	Normal dose	
Fluoroquinolones	Ceftriaxone		Normal dose	
		Ciprofloxacin	Reduce in severe impairment	PQ ₄ binders may impair absorption
		Norfloxacin	Reduce in severe impairment	PO ₄ binders may impair absorption
Glycopeptides	Moxifloxacin		Normal dose	
		Teicoplanin	Avoid/dose reduction	Nephrotoxicity and ototoxicity
		Vancomycin	Avoid/dose reduction	Nephrotoxicity and ototoxicity
Macrolides	Azithromycin		Normal dose	
	Clarithromycin		Normal dose	CYP3A4 interactions
	Roxithromycin		Normal dose	
Penicillins		Erythromycin	Dose reduction	CYP3A4 interactions
		Amoxycillin	Normal dose	
		Benzylpenicillin	Dose reduction	
		Piperacillin	Dose reduction	
		Ticarcillin/clavulanic acid	Dose reduction	
Sulfonamides and trimethoprim			Reduce high doses	Falsely elevates creatinine
Tetracyclines		Tetracycline	Avoid	
	Doxycycline		Normal dose	
	Minocycline		Normal dose	
Antimycobacterials		Ethambutol	Dose reduction	
	Isoniazid		Normal dose	
	Pyrazinamide		Normal dose	
Azole antifungals	Rifampicin		Normal dose	CYP3A4 interactions
		Fluconazole	Reduce maintenance doses	
	Ketoconazole		Normal dose	CYP3A4 interactions

Continued

TABLE 78.1 Common Drug Dosing Recommendations in Chronic Kidney Disease Including Kidney Transplantation and Dialysis—cont'd

Drugs	PREDOMINANT ELIMINATION PATHWAY		Dosing Recommendations	Comments
	Hepatic	Kidney		
Other antifungals	Itraconazole		Normal dose	CYP3A4 interactions
	Voriconazole		Normal dose	CYP3A4 interactions
		Terbinafine	Reduce maintenance doses	
Guanine analogs	Caspofungin		Normal dose	
	Griseofulvin		Normal dose	
		Acyclovir	Dose reduction	CNS + nephrotoxicity
Other antivirals		Famciclovir	Dose reduction	CNS + nephrotoxicity
		Ganciclovir	Dose reduction	Hematologic toxicity
		Valaciclovir	Dose reduction	CNS + nephrotoxicity
Neuraminidase inhibitors		Entecavir	Avoid/dose reduction	
		Ribavirin	Avoid/dose reduction	Hematologic toxicity
		Oseltamivir	Dose reduction	
Other antivirals		Cidofovir	Significant dose reduction	Nephrotoxic
		Foscarnet	Significant dose reduction	Nephrotoxic
Immunosuppressants				
Calcineurin inhibitors	Cyclosporin		Normal dose	Nephrotoxic
	Tacrolimus		Normal dose	Nephrotoxic
mTOR inhibitors	Everolimus		Normal dose	
	Sirolimus		Normal dose	
Antiproliferatives	Azathioprine		Normal dose	
	Mycophenolate		Normal dose	Hematologic toxicity
		Cyclophosphamide	Dose reduction	
Anticoagulants, Antiplatelet Agents, Thrombolytics, and Hemostatics				
UFH	UFH		Normal dose	
LMWHs		Dalteparin	Dose reduction	
		Enoxaparin	Dose reduction	Bleeding risk
Anticoagulants	Warfarin		Avoid/normal dose	May promote vascular calcification
		Apixaban	Avoid/dose reduction	Bleeding risk
		Bivalirudin	Avoid/dose reduction	Bleeding risk
		Dabigatran	Avoid/dose reduction	Bleeding risk
		Danaparoid	Avoid/dose reduction	Bleeding risk
		Fondaparinux	Avoid/dose reduction	Bleeding risk
		Lepirudin	Avoid/dose reduction	Bleeding risk
		Melagatran	Avoid/dose reduction	Bleeding risk
		Ximelagatran	Avoid/dose reduction	Bleeding risk
		Rivaroxaban	Avoid/dose reduction	Bleeding risk
Antiplatelets		Aspirin	Normal dose	
		Clopidogrel	Normal dose	
		Dipyridamole	Normal dose	
Cardiovascular Drugs				
ACE inhibitors			Titrate according to kidney function and serum potassium	Potential nephrotoxicity
ARBs				Potential hyperkalemia
β -Blockers		Atenolol	Dose reduction	
		Sotalol	Dose reduction	
	Carvedilol		Normal dose	
	Labetalol		Normal dose	
	Metoprolol		Normal dose	
	Pindolol		Normal dose	
	Propranolol		Normal dose	

TABLE 78.1 Common Drug Dosing Recommendations in Chronic Kidney Disease Including Kidney Transplantation and Dialysis—cont'd

Drugs	PREDOMINANT ELIMINATION PATHWAY		Dosing Recommendations	Comments
	Hepatic	Kidney		
CCB	Amlodipine Diltiazem Lercanidipine Nifedipine Verapamil		Normal dose Normal dose Normal dose Normal dose Normal dose	CYP3A4 interactions
Antianginals	Nicorandil Nitrates Perhexiline		Normal dose Normal dose Normal dose	CYP3A4 interactions
Statins	Atorvastatin Rosuvastatin Simvastatin		Normal dose Normal dose Normal dose	
Fibrates		Pravastatin Fenofibrate	Normal dose Dose reduction	
Antiarrhythmics		Digoxin Disopyramide Procainamide	Reduce dose + monitoring Dose reduction Dose reduction	Bradycardia
	Amiodarone Flecainide Mexiletine		Normal dose Normal dose Normal dose	
Diuretics		Furosemide Spironolactone	Dose increase Avoid	Hyperkalemia
Diabetes				
Insulins		Insulin	Cautious dose titrations	
Biguanides		Metformin	Avoid	Lactic acidosis
Sulfonylureas	Gliclazide Glibenclamide Glipizide		Normal doses Avoid/dose reduction Avoid/dose reduction	Active metabolites Active metabolites
DPP-4 inhibitors	Linagliptin		Normal dose	
		Saxagliptin Sitagliptin Vildagliptin	Avoid/dose reduction Dose reduction Dose reduction	
SGLT2 inhibitors		Dapagliflozin Empagliflozin Ertugliflozin	Avoid eGFR <45 mL/min/1.73 m ² Avoid eGFR <30 mL/min/1.73 m ² Avoid eGFR <45 mL/min/1.73 m ²	Associated with severe DKA
GLP-1 analogs		Exenatide	Avoid eGFR <30 mL/min/1.73 m ²	
	Dutaglutide Liraglutide Semaglutide			Normal dose Normal dose Normal dose
Thiazolidinediones	Pioglitazone Rosiglitazone		Avoid Avoid	
Gastrointestinal				
H ₂ antagonists		Ranitidine	Normal dose	
PPIs	Cimetidine Esomeprazole Lansoprazole Omeprazole Pantoprazole		Avoid Normal dose Normal dose Normal dose Normal dose	Drug interactions Acute interstitial nephritis may occur (see Chapter 64)

Continued

TABLE 78.1 Common Drug Dosing Recommendations in Chronic Kidney Disease Including Kidney Transplantation and Dialysis—cont'd

PREDOMINANT ELIMINATION PATHWAY				
Drugs	Hepatic	Kidney	Dosing Recommendations	Comments
Dopamine antagonists	Domperidone Metoclopramide Prochlorperazine		Normal dose Normal dose Normal dose	Extrapyramidal and CNS effects at high dose with CKD
5-HT ₃ antagonists	Ondansetron Dolasetron Tropisetron		Normal dose Normal dose Normal dose	
Aperients and laxatives	Low potency: Bisacodyl Docusate Lactulose Sorbitol High potency, including phosphate containing		Normal dose Normal dose Normal dose Normal dose Avoid	
Antidiarrheal agents	Atropine Diphenoxylate Loperamide		Normal dose Normal dose Normal dose	
Musculoskeletal				
Gout		Allopurinol	Reduce dose	Active metabolite Avoid in adynamic bone disease Rapid IV nephrotoxic
	Febuxostat		Normal dose	
Bisphosphonates		Alendronate	Normal or reduced dose	
		Pamidronate	Normal or reduced dose	
		Risedronate	Normal or reduced dose	
		Zoledronate		
CNS-Acting Drugs				
Antiepileptics		Levetiracetam	Reduce dose	CYP3A4 interactions CYP3A4 interactions
	Carbamazepine		Normal dose	
	Phenytoin		Normal dose + monitoring	
	Valproate		Normal dose	
SSRIs	Citalopram Paroxetine Venlafaxine		Normal dose Normal dose Normal dose	
Antipsychotics	Clozapine Olanzapine Risperidone		Normal dose Normal dose Normal dose	Enhanced CNS toxicity
Benzodiazepines	Clonazepam Diazepam Oxazepam Temazepam		Normal dose Normal dose Normal dose Normal dose	

ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CCB, calcium channel blocker; CKD, chronic kidney disease; CNS, central nervous system; DKA, diabetic ketoacidosis; DPP-4, dipeptidyl peptidase-4; eGFR, estimated glomerular filtration rate; GLP-1, glucagon-like peptide-1; H₂, histamine-2; IV, intravenous; LMWH, low-molecular-weight heparin; mTOR, mammalian target of rapamycin; NSAID, nonsteroidal antiinflammatory drug; PO₄, phosphate; PPI, proton pump inhibitor; SGLT2, sodium-glucose cotransporter-2; SSRI, selective serotonin reuptake inhibitor; TCA, tricyclic antidepressant; UFH, unfractionated heparin.

suppression, which is exacerbated by combined use with other bone marrow suppressants (e.g., ganciclovir, cotrimoxazole, and mTOR inhibitors). Regular blood monitoring is needed to guide dosage. Methotrexate relies on kidney clearance and should be avoided or used cautiously at a reduced dose in people with a low GFR. Kidney clearance is significant with cyclophosphamide, which can cause hemorrhagic cystitis via an active metabolite (acrolein) if it is allowed to dwell in the bladder for lengthy periods

(hence it should not be administered prior to bedtime), and chlorambucil, and dose modifications are required. Allopurinol and febuxostat significantly interfere with metabolism of the active metabolite of azathioprine (6-mercaptopurine), and coprescription can lead to life-threatening bone marrow suppression. The combination should be avoided by exchanging azathioprine with mycophenolate or by significant (75%) dose reduction of azathioprine or mercaptopurine.

mTOR Inhibitors

Although not nephrotoxic when used alone, mTOR inhibitors (sirolimus, everolimus) can potentiate the nephrotoxicity of CNIs. mTOR inhibitors are almost entirely hepatically metabolized to essentially inactive metabolites. Dosage is independent of kidney function and based on response, serum drug levels, and specific toxicities (especially hematologic and lipids). mTOR inhibitors are substrates of CYP3A4 and P-glycoprotein and susceptible to the same drug interactions as CNIs (Table 78.2). Bone marrow toxicity (neutropenia, anemia, and thrombocytopenia) is increased when they are used in combination with other myelosuppressants (azathioprine, mycophenolate, ganciclovir, and cotrimoxazole). Trough levels show good correlation with drug efficacy/toxicity and are used to monitor therapy. Different assay methodologies (HPLC vs. immunoassays) vary in their ability to detect parent drug from metabolites. Specific HPLC methods are recommended, and results from different laboratories or methods may not be interchangeable.

Immunosuppressant Antibodies

Antibodies used for immunosuppression include T cell–depleting antibodies (antithymocyte globulin, anti–interleukin-2 receptor (CD25) antibodies (basiliximab and daclizumab), and B cell–depleting antibodies (rituximab) (see Chapter 106). Dosage is independent of kidney function, and these agents should be administered after plasma exchange²⁸ to avoid drug removal. They are not removed by HD.

CHEMOTHERAPY AND CYTOTOXIC AGENTS

Kidney function affects the outcomes of patients with cancer due to the impact on the anticancer drug selection and dosing.²⁹ Alternative use of the second- or third-line agent may lead to reduced effectiveness; underestimation of kidney function may lead to inappropriate agent selection, dose reductions, underdosing, reduced effectiveness,

and failure of therapy. Conversely, overestimation of kidney function may lead to overdosing and toxicity from anticancer drugs cleared by the kidney. Cytotoxic agents are drugs with a narrow therapeutic index, hence a small change in plasma concentrations can lead to toxicity or loss of efficacy. Many anticancer drugs are excreted predominantly as unchanged drug or active metabolite in the urine and therefore require dose adjustment for kidney function (e.g., bleomycin, capecitabine, carboplatin, cisplatin, cyclophosphamide, erlotinib, etoposide, fludarabine, imatinib, methotrexate, mitomycin, pemetrexed).

Carboplatin dose is calculated based on the Culvert equation, which takes into consideration the target AUC and the patient's kidney function:

$$\text{Dose (mg)} = \text{Target AUC} \times (\text{GFR} + 25)$$

Carboplatin dosing varies significantly on the estimate of GFR. The dose depends on the method used for estimation of GFR and whether a body surface area (BSA) indexed eGFR or absolute eGFR is used in the Culvert formula.

Tyrosine Kinase Inhibitors

Tyrosine kinase inhibitors are novel small molecule targeted therapies administered orally and associated with improved outcomes and reduced toxicity compared with the conventional chemotherapy. Most of these agents do not require dose adjustment in mild and moderate CKD.³⁰ The pharmacokinetics for some of these agents have not been studied in patients with severe CKD, and therefore their use has not been recommended in this patient population (e.g., alectinib, cabozantinib) or may be administered with caution (e.g., ceritinib, dabrafenib, trametinib).

Some of these agents require dose adjustment in patients with severe CKD (e.g., afatinib, brigatinib, crizotinib, vandetanib). Crizotinib is an anaplastic lymphoma kinase 1 (ALK-1) inhibitor. It may act as a

TABLE 78.2 Common CYP3A4/P-gp–Mediated Interactions With Transplant Immunosuppressants (Cyclosporine, Tacrolimus, Sirolimus, Everolimus)

Drug Class	Examples	Relative Potency
Drugs That Inhibit CYP3A4 and P-gp		
Macrolide antibiotics	Azithromycin	–
	Clarithromycin	+++
	Erythromycin	+++
	Roxithromycin	+
	Fluconazole	+/-
Azole antifungals	Itraconazole	++
	Ketoconazole	+++
	Voriconazole	+++
	Felodipine	++
CCBs	Amlodipine	+
	Nifedipine	+/-
	Felodipine	+/-
	Diltiazem	++
	Verapamil	+++
Herbals, foods	Grapefruit juice	++
Drugs That Induce CYP3A4 and P-gp		
Antiepileptics	Carbamazepine	++
	Phenytoin	+++
	Phenobarbitone	++
Antibiotics	Rifampin	+++
Herbals, foods	St. John's wort	+++

CCB, Calcium channel blocker; P-gp, P-glycoprotein.

competitive inhibitor of CrCl by the organic cation transporter 2 (OCT2) in the proximal tubule. This effect is similar to other drugs (e.g., trimethoprim, cimetidine), leading to the increase in serum creatinine that is not associated with true AKI (i.e., pseudo-AKI).

ANTICOAGULANTS, ANTIPLATELET AGENTS, THROMBOLYTICS, AND HEMOSTATICS

Bleeding risk with anticoagulants, antiplatelet agents, and thrombolytics is generally increased in patients with CKD.³¹

Unfractionated Heparin

Unfractionated heparin (UFH) is cleared via a saturable mechanism of the endothelium and reticuloendothelial system and a nonsaturable mechanism through the kidneys. Half-life is slightly prolonged (1.5-fold) in CKD, especially at higher doses. However, UFH is used at normal dose with titration based on activated partial thromboplastin time monitoring. In some centers, UFH is preferred in severe CKD because its duration of effect is shorter and more easily measured and reversed.

Low-Molecular-Weight Heparins

Low-molecular-weight heparins (LMWHs) (enoxaparin, dalteparin) rely more on kidney clearance than UFH and in moderate to severe CKD can accumulate, increasing the risk for serious bleeding, especially following surgery. Consideration should be given to using UFH in this setting. In situations where LMWHs are preferred and there is doubt about clearance (or other kinetic issues), dosing should be guided by anti-Xa activity assay. Tinzaparin does not accumulate in patients with CrCl greater than 20 mL/min after at least 10 days of dosing and may be a useful option for bridging anticoagulation.

Other Parenteral Anticoagulants

Other parenteral anticoagulants that accumulate in CKD and that should be avoided or used with significant dose reduction include bivalirudin, danaparoid, fondaparinux, lepirudin, ximelagatran, and melagatran. Argatroban may be used at normal dose. Prostacyclin, used to prevent platelet aggregation in HD circuits, is rapidly hydrolyzed and not affected by CKD.

Oral Anticoagulants

The liver extensively metabolizes warfarin, and in CKD it is usually commenced at normal dose and titrated according to the international normalized ratio. Warfarin is highly protein bound, and with hypoalbuminemia there may be increased sensitivity to warfarin. Similarly, hepatic metabolism may be altered with CKD that results in increasing sensitivity. Warfarin may promote vascular calcification and should be avoided in advanced CKD unless absolutely necessary (e.g., those with a mechanical heart valve). Newer, direct-acting oral anticoagulants (DOACs) (rivaroxaban and apixaban) are partially dependent on kidney clearance. Dabigatran is very dependent on kidney clearance. Because these agents are difficult to monitor (assays not routinely available), they are dose reduced according to estimates of CrCl and are contraindicated (rivaroxaban <15 mL/min, apixaban <25 mL/min, and dabigatran <30 mL/min) in advanced CKD.

Antiplatelet Drugs

Commonly prescribed doses of oral antiplatelets (aspirin, clopidogrel, and dipyridamole) do not require adjustment, although patients with CKD are more prone to bleeding. Despite their efficacy in reduced

kidney function, eptifibatide and tirofiban are excreted by the kidney and have been associated with bleeding in this setting. Abciximab is cleared by platelet binding and is not associated with increased bleeding risk in patients with CKD.

Thrombolytics

Streptokinase, anistreplase, and alteplase are used as normal, but the high risk for hemorrhage should be considered. Urokinase and alteplase are also used to unclot dialysis catheters and occasionally to prevent catheter dysfunction.

Hemostatics

Tranexamic acid is highly (90%) dependent on clearance via the kidneys and requires dose reduction in moderate to severe CKD. Protamine and vitamin K are used as normal, as are fresh-frozen plasma and whole blood with critical bleeding.

DIURETICS

Diuretics must reach the tubular lumen of the kidney in unbound form to exert an effect.³² Pharmacokinetic and pharmacodynamic properties of diuretics can change with proteinuria or CKD, usually causing a resistance to their effect. Diuretics in the immediate post-transplant setting may increase urine flow in native kidneys, giving a false impression of transplant function.

Thiazide Diuretics

Thiazide diuretics (chlorthalidone, hydrochlorothiazide, and indapamide) become less effective as diuretics when GFR is less than 30 mL/min/1.73 m², although they may augment the effectiveness of a loop diuretic and retain their antihypertensive effects. Metolazone can maintain efficacy at a lower GFR.

Loop Diuretics

Loop diuretics (furosemide, bumetanide, torsemide, and ethacrynic acid) remain effective at low GFR and are generally preferred in patients with CKD, although higher doses are usually required because less reaches the site of action in the lumen as GFR falls.

Potassium-Sparing Diuretics

Potassium-sparing diuretics are the least effective diuretics and should be used cautiously in moderate to severe CKD because of the risk for hyperkalemia.

ANTIHYPERTENSIVES

Antihypertensives should be used cautiously to avoid kidney hypoperfusion.³³ Hypotension during HD may require that doses be withheld or delayed on dialysis days.

ACE Inhibitors and ARBs

Although various ARBs and the active metabolites of ACE inhibitors are excreted by the kidney, they can be used effectively in CKD if initiated at moderate doses and titrated to response, kidney function, and serum potassium level. An increase in serum creatinine (up to 30% in the first 2 months) may be associated with blood pressure response. All ACE inhibitors and ARBs can cause AKI by inhibiting angiotensin II-mediated homeostatic vasoconstriction of the efferent renal arteriole during kidney hypoperfusion (e.g., with dehydration, hypotension, blood loss, and infection) or preexisting CKD (see also [Chapter 70](#)). Nephrotoxicity is more likely with coadministration of drugs that reduce kidney perfusion, including diuretics, antihypertensives, and NSAIDs.³⁴ Hyperkalemia is more likely in those with CKD and those

taking potassium-sparing diuretics or supplements. ACE inhibitors or ARBs also can worsen anemia by reducing erythropoietin production.

β-Blockers

Most β-blockers (carvedilol, labetalol, metoprolol, pindolol, and propranolol) are hepatically metabolized and used at normal doses. Sotalol and atenolol are 90% removed and bisoprolol 50% removed by the kidney, and so dose reductions are often but not always required as they can be dosed to clinical effect. Atenolol can sometimes be given three times weekly after dialysis. β-Blockers may slightly elevate serum potassium.

Calcium Channel Blockers

Pharmacokinetics of CCBs are generally unaltered in CKD. These drugs are usually well tolerated and used in normal doses. Dihydropyridine CCBs (especially amlodipine) can cause edema in the lower limbs that is more likely due to “capillary leakage” than fluid retention, and it is less responsive to diuretics. This can aggravate the edema of CKD. To varying extents, CCBs inhibit CYP3A4 and P-glycoprotein, causing increased absorption and reduced elimination of various substrate drugs, including CNIs and mTORs. Verapamil and diltiazem are moderately potent inhibitors, and dose modification of the affected drug is required. This interaction has been exploited so that diltiazem and other, more potent CYP3A4 and P-glycoprotein inhibitors are used as immunosuppressant-sparing agents.^{35,36} Other CCBs do not usually cause clinically significant interactions.

Other Antihypertensives

Methyldopa, clonidine, prazosin, terazosin, doxazosin, and minoxidil can be initiated and titrated at conventional dosage but may be associated with a higher incidence of adverse effects in patients with CKD. In particular, minoxidil is relatively contraindicated in CKD because it can lead to volume expansion with the development of pericardial and pleural effusions; when used, additional diuretic therapy may be required. α-Blockers cause profound orthostatic hypotension. Nitroprusside must be used cautiously because the toxic metabolite thiocyanate may accumulate with CKD but can be removed by hemodialysis.

ANTIANGINAL AGENTS

Most antianginals (nitrates, CCBs, nicorandil, and perhexiline) can be used as normal.

ANTIARRHYTHMICS

Various antiarrhythmics (digoxin, flecainide, disopyramide, procainamide, and sotalol) rely on kidney excretion and require dose modification. Digoxin has significant kidney excretion and a narrow therapeutic window, so dose reduction is essential even in mild CKD. Because of reduced tissue protein binding and V_D , some physicians use a smaller loading dose of digoxin than in patients with normal kidney function. Cautious monitoring and titration can prevent accumulation and toxicity. When possible, monitoring of antiarrhythmic drug concentrations and the electrocardiogram is recommended. Other agents (amiodarone, flecainide, metoprolol, mexiletine, and verapamil) are used at normal doses.

LIPID-LOWERING AGENTS

Bile Acid–Binding Resins

Bile acid–binding resins are now rarely used, and the large fluid volumes required to administer them limit their use in fluid-restricted

kidney patients. They can interfere with absorption and enterohepatic recirculation of various drugs, including mycophenolate.

Statins

Most statins (hydroxymethylglutaryl–acetyl CoA [HMG–CoA] reductase inhibitors) are extensively metabolized and can be used effectively at normal doses in CKD and those who have undergone transplantation.³⁷ Pravastatin is partially cleared via the kidneys. Rhabdomyolysis with AKI can occur, although the risk does not appear to be greatly increased with CKD. The risk increases with concomitant use of fibrates and drugs that inhibit the CYP3A4 metabolism of most statins. Patients commencing therapy should have kidney function and creatine kinase monitored regularly.

Fibrates

Fenofibrate but not gemfibrozil has extensive kidney clearance, and dose reduction is required. Combination of statins and fibrates significantly increases the risk for rhabdomyolysis and should be used only when benefits outweigh risks and with monitoring for muscle symptoms, creatine kinase, and alanine aminotransferase.

DIABETES

In addition to diabetes being a cause of CKD, the kidney also plays an important role in insulin metabolism, and thus kidney function influences glycemic control. Diabetes is common after transplantation because of resumed insulin metabolism by the functioning transplant and also the effect of tacrolimus and corticosteroids. Patients with CKD are at increased risk for hypoglycemia, and drugs should be initiated and titrated cautiously.³⁸ Various antidiabetic drugs depend on kidney excretion, and accumulation in CKD can cause adverse effects.³⁹ Management of diabetes in CKD is discussed in [Chapter 33](#).

Diabetes Management in Peritoneal Dialysis

Patients receiving PD may have higher antihyperglycemic requirements because of the glucose load in PD fluid. If insulin is administered intraperitoneally, dosages may differ from intravenous requirements. Icodextrin solutions can significantly interfere with blood glucose monitoring because of metabolites (maltose, maltotriose, or maltotetrose) that falsely elevate blood glucose readings when monitors that use the enzyme glucose dehydrogenase pyrroloquinoline quinone reaction are used. This may lead to administration of insulin and the development of life-threatening hypoglycemia. A glucose-specific test strip is required to avoid interference.

Biguanides

Metformin is excreted almost entirely unchanged in urine, and accumulation can contribute to severe or fatal lactic acidosis (see [Chapter 33](#)),⁴⁰ although the association is inconsistent. Metformin should be temporarily discontinued in situations known to increase the risk for lactic acidosis or reduce kidney function (e.g., acute tissue hypoxia, dehydration, serious infection, or trauma) and 24 to 48 hours before anticipated surgery or use of iodinated radiocontrast media. Current Kidney Disease: Improving Global Outcomes guidelines recommend metformin as first-line therapy in CKD but to discontinue it where GFR is less than 30 mL/min/1.73 m². Even with low-dose use in patients with GFR less than 30 mL/min/1.73 m², there is still a risk for lactic acidosis, and an acute deterioration in kidney function can reduce drug clearance at any time. Patients on this regimen should be advised to seek early medical advice in any cases of acute deterioration in health.

Insulins

As GFR decreases, insulin clearance decreases in parallel, and so hypoglycemia may ensue if kidney function deteriorates and insulin regimens are not adjusted accordingly. Uremia, however, can cause peripheral resistance to insulin, requiring increased doses. Most insulin regimens can be used in patients with CKD or transplantation with cautious titration and monitoring.

Meglitinides

The nonsulfonylurea insulin secretagogues repaglinide and nateglinide can be used in patients with kidney failure without dose adjustment.

Sulfonylureas

Sulfonylureas are metabolized, and some (glibenclamide and glimepiride) have active metabolites that are excreted by the kidney. In moderate to severe CKD, the risk of hypoglycemia is increased.⁴¹ These agents should be initiated at a low dose and titrated to response. Gliclazide and glipizide are preferable because they do not have active metabolites. Regardless of which agent is used, the effect of sulfonylureas still may be increased because the insulin they release will itself have a prolonged action in CKD.

Sodium-Glucose Cotransporter-2 Inhibitors

Sodium-glucose cotransporter-2 inhibitors (e.g., dapagliflozin, canagliflozin, empagliflozin) depend on adequate kidney function to lower blood glucose levels and may be less effective at achieving this outcome in moderate or severe CKD. Nevertheless, trials suggest their independent renoprotective benefits by reducing proteinuria and delaying diabetic kidney disease.^{42,43} Dapagliflozin, empagliflozin, and ertugliflozin are currently not licensed for eGFR less than 45 mL/min/1.73 m².

Dipeptidyl Peptidase-4 Inhibitors

Many dipeptidyl peptidase-4 inhibitors (saxagliptin, sitagliptin, vildagliptin) rely on kidney excretion and require significant dose reduction in CKD. No dose adjustment is required for linagliptin.

Glucagon-Like Peptide-1 Receptor Agonists

Several glucagon-like peptide 1 analogs (dulaglutide, liraglutide, semaglutide) may be used without dose adjustment even in severe CKD. However, exenatide should be used with caution when eGFR is less than 45 mL/min/1.73 m² and is not recommended when eGFR is less than 30 mL/min/1.73 m².

Thiazolidinediones

Thiazolidinediones are extensively metabolized and excreted in bile. Their pharmacokinetics are not significantly altered by CKD and in fact show reduced exposure, possibly because of reduced protein binding. However, they can cause fluid retention and edema, exacerbating the difficulties of fluid management and heart failure and are generally best avoided in CKD. Thiazolidinediones have been associated with dilutional anemia because of an increase in plasma volume, which may complicate management of the anemia associated with CKD.

DRUGS FOR THYROID DISORDERS

Thyroid hormones generally do not require dose alteration in CKD.⁴⁴ Doses should be initiated and titrated to thyroid-stimulating hormone levels and clinical effect. In circulation, thyroxine (T₄) is 99.98% protein bound (0.02% free), and triiodothyronine (T₃) is 99.8% bound (0.2% free). T₃ and T₄ bind partially, in slightly different proportions, to three different plasma proteins: thyroid-binding globulin, thyroid-binding prealbumin, and albumin. In protein-deficient states (e.g., nephrotic syndrome), there is the possibility of transient or permanent changes in thyroid hormone protein binding that may alter the free fraction of T₃

and T₄, leading to transient toxicity. Uremic toxins can inhibit enzymes associated with conversion of T₄ to T₃. Oral absorption of thyroid hormones is affected by coadministration with metallic phosphate binders and iron. Euthyroid patients with CKD may have abnormal thyroid function test results, possibly because of decreased peripheral conversion of T₄ to T₃, decreased clearance of reverse T₃ generated from T₄, or decreased binding of thyroid hormones to proteins. Antithyroid (carbimazole and propylthiouracil) can be used at usual doses.

MINERAL AND BONE DISORDERS

Prescribing for these disorders is further discussed in [Chapter 88](#).

Phosphate Binders

Phosphate binders should be taken with meals for maximal efficacy. Patients can be educated to adjust phosphate binder dose to the phosphate content in each meal. Doses are not affected by kidney function, except that the requirement for phosphate binders tends to increase as GFR declines. Dosage is based on plasma phosphate concentration and the need to avoid biochemical abnormalities. Acid suppression may reduce the effectiveness of some phosphate binders but not sucralfate. Phosphate binders also may reduce gastrointestinal absorption of drugs, including thyroid hormones, fluoroquinolones, tetracyclines, digoxin, and immunosuppressants.

Vitamin D

In patients with CKD, inability of the kidneys to convert 25-hydroxycholecalciferol to the active 1,25 dihydroxycholecalciferol (calcitriol) may produce relative vitamin D deficiency and hypocalcemia, which requires treatment with the active vitamin D preparations calcitriol, paricalcitol, or alfacalcidol.

Calcimimetics

Cinacalcet dosage is independent of kidney function. When possible, cinacalcet should be administered with the evening meal to improve absorption and minimize side effects. Similarly, morning blood samples for parathyroid hormone levels should be drawn at least 12 hours after administration. Cinacalcet and etelcalcetide (a calcimimetic for parenteral use), are further discussed in [Chapter 90](#).

DYSPEPSIA, GASTROESOPHAGEAL REFLUX DISEASE, AND PEPTIC ULCERS

Proton pump inhibitors, histamine-2 (H₂) receptor blockers, and antacids are commonly used in patients with CKD (see [Chapter 90](#)).

Antacids

Alginates, magnesium trisilicate, and sodium bicarbonate are useful for symptom control, but the latter has high sodium content, which exacerbates hypertension and fluid status. Magnesium and aluminum salts can reduce the absorption of mycophenolate. Aluminum salts were widely used in the past as phosphate binders but are now often avoided in patients with advanced CKD due to the toxicity from absorbed aluminium (especially encephalopathy and bone toxicity).

H₂ Antagonists

Most H₂ antagonists are removed by the kidney but are relatively safe, so dose reduction is usually not required.⁴⁵ Protein binding and V_D are unaltered; however, the bioavailability of nizatidine is reduced in CKD. Cimetidine should be avoided because accumulation produces CNS effects, and it causes significant interactions through the CYP450 system and falsely elevates serum creatinine levels.

Proton Pump Inhibitors

Proton pump inhibitors (PPIs) are generally safe and well tolerated even at normal doses, although they have been associated with interstitial nephritis as well as with increased risk for CKD. As such, subjects with CKD should be evaluated for their need of PPIs or switched to H₂ antagonists if possible.

ANTIEMETICS

Dopamine Antagonists

Domperidone, metoclopramide, and prochlorperazine are not significantly removed by the kidney; however, extrapyramidal and CNS effects may occur in CKD as well as QT prolongation, especially at high doses. Domperidone does not cross the blood-brain barrier and may be preferable for long-term management. Metoclopramide and domperidone increase gastric emptying, which may alter drug pharmacokinetics.

5-HT₃ Antagonists

Most 5-HT₃ antagonists are minimally excreted in urine and have a wide therapeutic window. Dolasetron, granisetron, ondansetron, and tropisetron are safe, and dose modification is not required.

APERIENTS AND LAXATIVES

Adequate fluid intake is important for normal bowel function, and fluid restrictions required in many end-stage kidney disease (ESKD) and dialysis patients are frequently accompanied by constipation as a consequence. Common low-potency agents, including docusate, bisacodyl, glycerin, lactulose, liquid paraffin, senna, and sorbitol, may be used for acute or chronic constipation. Normal doses should be titrated to effect while avoiding significant dehydration, fluid shifts, or electrolyte disturbances. High-potency laxatives and bowel preparations should be used cautiously, as they may cause significant fluid and electrolyte disturbances. Preparations containing high amounts of phosphate should be avoided. Despite the large fluid volumes required for administration, isoosmotic laxatives may be used for bowel preparation in patients with CKD and dialysis.

ANTIDIARRHEALS

Opioids or their derivatives should be used with the same caution as when they are used as analgesics. Loperamide and diphenoxylate-atropine can be given in usual doses.

DRUGS FOR ERECTILE DYSFUNCTION

Phosphodiesterase-5 Inhibitors

The AUCs of sildenafil (2-fold), tadalafil (4-fold), and vardenafil (20%–30%) are increased in patients with severe CKD despite minimal dependence on kidney excretion. However, provided that relevant CV and drug contraindications are excluded, they can be used safely. They should be initiated at low doses and titrated to response. Shorter-acting agents (sildenafil and vardenafil) may be preferable. Vardenafil can prolong the QT interval.

Intracavernosal Therapy

Drugs given directly by intracavernosal injection do not achieve significant concentrations in the systemic circulation and can be used in patients with CKD.

MUSCULOSKELETAL DRUGS

Nonsteroidal Antiinflammatory Drugs

NSAIDs can cause significant nephrotoxicity and should be avoided or used with extreme caution in patients with CKD.

Miscellaneous Arthritis Drugs

Gold salts and penicillamine are now rarely used for rheumatoid arthritis. Both were associated with nephrotic syndrome caused by membranous nephropathy. Glucosamine and fish oil can be used safely.

Gout and Hyperuricemia

Gout is common in CKD due to the impaired ability to remove urate. In acute therapy, short courses of oral corticosteroids are safe and preferable to NSAIDs. Colchicine accumulation in CKD may cause diarrhea and hypoperfusion-induced CKD, as well as myelosuppression, and dosing should be reduced. In maintenance therapy, allopurinol is effective but should be initiated in low doses (50–100 mg/day) and then slowly increased over weeks, as initiation with higher disease has been associated with the allopurinol hypersensitivity syndrome (a Stevens-Johnson–like syndrome). The toxicity is greater in subjects who carry the HLA-B*58:01 allele, which is more common in Chinese (Han) and Korean peoples. Oxypurinol (the active metabolite) can also accumulate in patients with low GFR.⁴⁶ Despite this, some patients with ESKD require and tolerate normal doses. Febuxostat is another xanthine oxidase inhibitor that is not affected by decreased kidney function but was associated with a higher risk for CV mortality than allopurinol in the CARES trial, although more recent studies did not confirm this finding. Allopurinol and febuxostat significantly interacts with azathioprine with a risk for severe myelosuppression. If the combination is unavoidable, the azathioprine dose should be reduced by 75% and blood counts monitored carefully. Uricosuric agents (e.g., probenecid) inhibit secretion of acids in the proximal tubule and prevent reabsorption of uric acid from the tubular lumen. They often become ineffective with diminishing kidney function and are best avoided if GFR is less than 40 mL/min/1.73 m². Probenecid also interferes with the tubular secretion of many drugs, thereby increasing plasma concentrations.

Bisphosphonates

Bisphosphonates are extensively excreted in urine. The fraction not excreted is incorporated into bone, from which it slowly dissociates. In CKD, impaired clearance of the absorbed drug may increase the fraction available for incorporation into bone. The long terminal elimination half-life of these agents reflects rate-limiting dissociation from bone and may not be significantly altered in patients with CKD. Oral bisphosphonates appear to be safe in CKD stages 2 and 3. Their safety in CKD stages 4 and 5 is less well established, and they are contraindicated in adynamic bone disease. Rapid administration of intravenous bisphosphonates (pamidronate and zoledronic acid) without hydration has been associated with acute nephrotoxicity. Intravenous preparations should be administered slowly with hydration and kidney function assessed regularly. Oral bisphosphonate administration may be complicated by the volumes of fluid recommended. In addition, many patients with CKD are taking calcium-based phosphate binders or supplements that impair the absorption of oral bisphosphonates.

ANTIEPILEPTICS

Patients with CKD may be more prone to seizures (e.g., uremic encephalopathy and dialysis disequilibrium syndrome) and to the CNS effects of antiepileptics. Some antiepileptics (especially gabapentin and

levetiracetam) are primarily cleared via the kidneys, and dose modification is essential. Some antiepileptics (barbiturates, phenytoin, and carbamazepine) are strong inducers of drug-metabolizing enzymes. Coadministration with drugs reliant on hepatic metabolism (e.g., immunosuppressants) can reduce exposure and efficacy of the concomitant drug. Therapeutic drug monitoring is available for many antiepileptics and should be used to guide dosage.

Benzodiazepines

See the “Psychotropic Drugs” section.

Carbamazepine

Carbamazepine is administered as normal and titrated to response and blood concentrations. It is a potent enzyme inducer, and care must be taken to account for important drug interactions.

Phenytoin

Caution should be exercised with phenytoin in CKD because of its erratic absorption, saturable metabolism, nonlinear pharmacokinetics, reduced protein binding, and increased V_D . The concentration of free drug may be higher than in normal kidney function due to displacement from protein binding sites by uremic toxins. Ideally, the unbound (active) concentration of phenytoin should be assayed. A low total serum phenytoin level in CKD should not be mistaken as subtherapeutic and a reason to increase the dose. Nystagmus, cerebellar ataxia, and seizures can occur in overdose. Cautious titration based on effect and monitoring of free concentration is advised. Phenytoin is a potent enzyme-inducing agent, and care must be taken to account for drug interactions.

Other Antiepileptics

Levetiracetam, topiramate, and vigabatrin undergo significant kidney excretion, and dose modification is required. Valproate and lamotrigine are not significantly excreted by the kidney and do not cause enzyme induction or inhibition. Protein binding changes can increase the free fraction of valproic acid, predisposing to increased responsiveness.

ANTIPARKINSONIAN DRUGS

In addition to their use for Parkinson disease and hyperprolactinemia disorders, dopaminergic drugs are used to treat restless legs and other limb movement disorders in patients with CKD. Most are hepatically cleared and safe, although dopaminergic agents may exacerbate postural hypotension. Amantadine is highly dependent on kidney excretion, and dose modification is essential.

ANTIMIGRAINE DRUGS

Simple analgesics (acetaminophen) are used as normal, although aspirin should be avoided at analgesic doses and opioids used cautiously (see discussion of analgesics). NSAIDs are best avoided because of potential nephrotoxicity. 5-HT₁ agonists are effective in patients with CKD and transplantation. Naratriptan relies the most on kidney excretion (50%), and a lower maximum dose is recommended. Sumatriptan and zolmitriptan are preferred because they are less dependent on kidney excretion.

PSYCHOTROPIC DRUGS

Most psychotropics are fat soluble and nondialyzable, undergo significant hepatic metabolism, and are excreted as inactive compounds. Even so, kidney patients are often more susceptible to common adverse

effects (especially CNS effects).⁴⁷ Slow titration and dose modifications are required. Adverse effects are not easily distinguished from symptoms of uremia. Despite being metabolized extensively by CYP450 enzymes, very few psychotropics cause clinically significant inhibition or induction of CYP3A4.

Selective Serotonin Reuptake Inhibitors

Most selective serotonin reuptake inhibitors (SSRIs) are extensively metabolized to compounds without significant SSRI activity.⁴⁸ Normal doses of citalopram, fluvoxamine, paroxetine, and sertraline can be used with cautious dose titration. Fluoxetine is metabolized to an active metabolite (norfluoxetine), but single- and multiple-dose studies have shown little change in pharmacokinetics, even in dialysis patients. Patients with CKD may be more prone to the CNS toxicity of SSRIs and serotonin syndrome. SSRIs can cause the syndrome of inappropriate antidiuretic hormone secretion and platelet aggregation or hemorrhagic complications, which may increase the risk for bleeding in uremic patients who are already prone to bleeding complications. Although most SSRIs have some inhibitory effect on CYP3A4, this is usually not significant. Fluoxetine and fluvoxamine are more likely to interact than citalopram and sertraline, but all can be used in patients taking immunosuppressants. Cinacalcet can cause significant inhibition of CYP2D6-mediated metabolism of SSRIs and perhexiline. This may increase the possibility of serotonin syndrome or perhexiline toxicity, though this risk can be mitigated by extra monitoring.

Tricyclic Antidepressants

Tricyclic antidepressants (TCAs) are now infrequently used for depression but more for neuropathic pain and sleep, and their anticholinergic properties in urinary tract disorders. They are predominantly metabolized to metabolites with varying activity. Patients with CKD may be more prone to the common anticholinergic adverse effects, particularly urinary retention, orthostatic hypotension, confusion, and sedation.⁴⁹ Some including amitriptyline can also prolong QT interval. Obstructive uropathy can occur from the anticholinergic properties. Cautious initial doses and titration of TCAs are recommended. Depending on response, many agents can be used at up to normal or maximal doses.

Monoamine Oxidase Inhibitors

Monoamine oxidase inhibitors (MAOIs) are extensively metabolized by the liver. Although they are now rarely used, normal doses of reversible MAOIs (moclobemide) are preferred and can be titrated cautiously to full doses. MAOIs can cause peripheral edema, which is not usually associated with fluid retention and is unresponsive to diuretics. Prostatic hypertrophy and urinary retention also may occur.

Other Antidepressants

Venlafaxine dose should be initially reduced in patients with severe CKD because of reduced clearance of the active metabolite O-desmethylvenlafaxine and the potential for hypertension. Nefazodone is used at a normal dose.

Antipsychotics

Conventional antipsychotics cause a variety of side effects to which patients with CKD may be susceptible (sedation, confusion, and postural hypotension). Caution is advised for atypical antipsychotics that prolong the QT interval (e.g., pimozide, thioridazine, mesoridazine, droperidol, and ziprasidone). Newer atypical agents are more commonly used and better tolerated. Clozapine, olanzapine, quetiapine, and aripiprazole are commenced at normal doses and titrated to response. The clearances of risperidone and its active metabolite 9-hydroxyrisperidone are reduced by 60% in severe CKD. Lithium is

filtered and reabsorbed mainly in the proximal tubule. It has extensive kidney clearance and accumulates even in mild CKD, causing toxicity, and should be used with increased plasma monitoring if required. Lithium is handled by the body like sodium, and any drug that affects plasma sodium concentration will likely affect plasma lithium concentration. Such drugs include diuretics that increase sodium loss, and the kidney “compensates” by retaining lithium and NSAIDs. HD is efficient in removal of lithium and can be used in overdose; however, multiple dialysis treatments are usually required because plasma concentrations rebound soon after HD. Chronic lithium nephrotoxicity is discussed further in [Chapter 65](#).

Benzodiazepines

Benzodiazepines are extensively metabolized by the liver to a range of active and inactive metabolites. Enhanced CNS toxicity, especially sedation, is the main concern in kidney patients. Because of the potential for accumulation, chronic use should be discouraged. Shorter-acting benzodiazepines are preferred, and the dose should be titrated cautiously according to response. The dose of midazolam should be reduced because of changes in plasma protein binding. Hemoperfusion and dialysis are not useful in patients with benzodiazepine intoxication. Flumazenil may be used as an antidote in patients with overdose, at doses similar to those with normal kidney function.

ANEMIA DRUGS

Erythropoiesis-Stimulating Proteins

The pharmacokinetics of erythropoiesis-stimulating agents (ESAs) are not affected by kidney function *per se*. However, as kidney function decreases, dose requirements may increase to account for reduced endogenous erythropoietin production. Use of ESAs is discussed in detail in [Chapter 86](#).

Iron Therapy

Oral supplements may be sufficient in early-stage CKD or PD patients who do not have the same degree of regular blood loss as HD patients. The most iron that can be absorbed from the gut is likely approximately 20 mg/day, and many iron tablets contain much more than this (generally 80–100 mg). Patients with severe iron deficiency

including those on HD often require intravenous supplementation (see [Chapter 86](#)).

ANTIHISTAMINES

Normal doses of sedating antihistamines generally can be used. They should be used cautiously in patients with bladder outflow obstruction because of their anticholinergic adverse effects that may cause or aggravate urinary frequency or retention. Newer, less sedating antihistamines are better tolerated and have a wider therapeutic window, and accumulation rarely causes significant complications. Cetirizine relies more on kidney clearance, and dose reduction is suggested. Loratadine and desloratadine have active metabolites but are safe. Fexofenadine is safe; however, terfenadine should be avoided because of the risk for arrhythmias.

VACCINES

Live vaccines (BCG [bacillus Calmette-Guérin], oral poliovirus, rubella, typhoid, yellow fever, and varicella) in immunosuppressed patients are contraindicated because of the potential for causing disease. Attenuated vaccines (diphtheria-tetanus, hepatitis B, influenza, meningococcal, and pneumococcal) may be used, but impaired response in immunocompromised individuals may lead to inadequate protection. Hepatitis B may need more doses for seroconversion to be achieved in patients on HD. Immunization should preferably occur at least 1 month before initiation of immunosuppression. After transplantation, the immune response may be inadequate for at least 6 to 8 months, meaning that vaccination should be withheld until then.⁵⁰

VITAMIN SUPPLEMENTATION

Patients with CKD may become vitamin deficient as a result of poor dietary intake and the effect of dialysis on removal of water-soluble vitamins. Administration of vitamin supplements (B, C, and folic acid) is recommended after dialysis. In contrast, liposoluble vitamin A (retinol, beta-carotene) may accumulate in patients on dialysis, causing hypervitaminosis A, and supplementation is not required.⁵¹

SELF-ASSESSMENT QUESTIONS

- Which of the following drugs require(s) clinically significant dose modification in patients with significantly reduced kidney function?
 - Enoxaparin
 - Gabapentin
 - Amoxicillin
 - Amlodipine
 - Famciclovir
- Which of the following drugs is/are the most likely to cause clinically significant drug interactions with calcineurin inhibitors?
 - Fluconazole
 - Clarithromycin
 - Vancomycin
 - Acyclovir
 - Roxithromycin
- Which of the following drugs require(s) dose modifications in chronic kidney disease because of a metabolite that is substantially removed by the kidney?
 - Olanzapine
 - Allopurinol
 - Atenolol
 - Amlodipine
 - Pethidine (meperidine)
- Which of the following drugs is/are likely to be removed by hemodialysis and should therefore be dosed after dialysis?
 - Pregabalin
 - Amiodarone
 - Levetiracetam
 - Pantoprazole
 - Acyclovir

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